Quade **Butler**

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Education

Doctor of Philosophy in Mechanical Engineering Dec 2025
McMaster University
Ontario Graduate Scholar, 2023 - 2024
Thesis: *Double Exponential Cubature Kalman Filtering: Theory and Applications*
Supervisor: *Dr. S. Andrew Gadsden*
GPA: *4.00/4.00*

Bachelor of Engineering in Mechanical Engineering 2020
University of Guelph
Minor in Mathematics
GPA: *3.80/4.00*

Experience

Research Associate Jan 2026 – Present
ICE Lab | McMaster University Hamilton, ON

- Advanced doctoral research on nonlinear Bayesian state estimation by refining double exponential cubature methods for Gaussian-weighted integration and probabilistic estimation
- Developed robust filtering algorithms for nonlinear dynamic systems with measurement outliers, improving estimator accuracy, consistency, and numerical stability

PhD Researcher Jan 2021 – Dec 2025
ICE Lab | McMaster University Hamilton, ON

- Developed double exponential cubature for multivariate Gaussian-weighted integration, deriving deterministic cubature rules for probabilistic estimation in nonlinear dynamic systems
- Introduced the Double Exponential Cubature Kalman Filter (DECKF) extending the cubature method into a recursive Bayesian state estimator and benchmarking it against UKF, CKF, and Gauss-Hermite filtering baselines
- Developed and evaluated robust nonlinear filtering methods for state estimation under outliers and model uncertainty using Monte Carlo simulation, RMSE, NEES/NIS, and statistical consistency testing
- Built an interacting multiple model fault detection and operating regime classification workflow for nonlinear time-series data, achieving 99.2% accuracy and 99.1% macro-F1 across induced operating regimes

ICE Lab | Research collaboration with Ford Motor Company Livonia, MI

- Collaborated with Ford and university research teams on a production-feasible condition monitoring strategy for machine tool health monitoring and maintenance planning
- Designed an industrial machine tool monitoring framework by balancing diagnostic value, sensor constraints, deployment cost, and production feasibility
- Built Python pipelines for profiling, cleaning, synchronizing, and feature engineering noisy industrial datasets, reducing measurement and analysis time by 30x through workflow redesign and degradation modeling

Undergraduate Researcher May 2019 – May 2020
University of Guelph Guelph, ON

- Studied strain partitioning in advanced high-strength automotive steels using in-situ SEM tensile testing and micro digital image correlation
- Prepared specimens, processed SEM image sequences, optimized DIC parameters, and contributed to peer-reviewed publication in Materials Characterization

Publications

2025	Gaussian Filtering Using a Spherical-Radial Double Exponential Cubature, Q. Butler, Y. Ziada, S. A. Gadsden	IEEE OJSP
2024	Strain partitioning characterization of advanced high strength steels using in-situ tensile tests with Micro digital image correlation – Methodology and analysis, A. Bardelcik, Q. Butler	Materials
2023	Preload Loss Detection in a Ball Screw System using Interacting Models, B. S. Sicard, Q. Butler, Y. Ziada, E. Hughey, S. A. Gadsden	IEEE OJIM
2022	Condition Monitoring of Machine Tool Feed Drives: A Review, Q. Butler, Y. Ziada, D. Stephenson, S. A. Gadsden	ASME JMSE

Conferences

2025	System identification and state estimation for magnetorheological dampers, P. Kosierb, Y. Wu, Q. Butler, B. Sicard, S. A. Gadsden	SPIE Signal Proc. XXXIV
2025	Friction Modeling and Monitoring for Machine Tool Health Management, B. Sicard, Y. Wu, Q. Butler, S. A. Gadsden	IEEE PHM
2024	Generating synthetic data for artificial intelligence solutions via a digital twin for condition monitoring in machine tools, B. Sicard, Q. Butler, Y. Wu, S. Abdolahi, Y. Ziada S. A. Gadsden	SPIE AI & ML II
2024	Nonlinear Filtering Using the Double Exponential Transformation, Q. Butler, B. Sicard, W. Hilal, S. A. Gadsden, Y. Ziada	SPIE Signal Proc. XXXIII
2023	Design Considerations for Building an IoT Enabled Digital Twin Machine Tool Sub-System, B. Sicard, Q. Butler, P. Kosierb, Y. Wu, Y. Ziada, S. A. Gadsden	IEEE AIBThings
2023	IIODT: Industrial Internet of Digital Twins for Hierarchical Asset Management in Manufacturing Sub-System, B. Sicard, Q. Butler, P. Kosierb, Y. Wu, Y. Ziada, S. A. Gadsden	IEEE AIBThings
2023	Generalizing the Unscented Kalman Filter for State Estimation, Q. Butler, W. Hilal, B. Sicard, Y. Ziada, S. A. Gadsden	SPIE Signal Proc. XXXII
2023	Cognitive dynamic digital twin: enhancements for digital twin platforms based on human cognition, B. Sicard, Q. Butler, Y. Ziada, S. A. Gadsden	SPIE Big Data V
2023	Experimental Setups for Linear Feed Drive Predictive Maintenance: A Review, B. Sicard, Q. Butler, Y. Ziada, S. A. Gadsden	IEEE PHM
2022	Rapid Parameter Estimation and Monitoring of CNC Feed Drive Systems, Q. Butler, B. Sicard, E. Hughey, Y. Ziada, D. Stephenson, S. A. Gadsden	SPIE Signal Proc. XXXI

Teaching Assistantships

2024 **Mechanical Vibrations**, McMaster University
2024 **Engineering Design**, McMaster University
2022 **Mechatronics**, McMaster University
2022 **Engineering Measurements**, McMaster University
2020, 2021 **Differential Equations**, University of Guelph

Academic Services

2025 **Reviewer**, ASME Journal of Tribology